

Lucy Robinson

Mechanical Engineering Student (MEng) • Themed Entertainment & Ride Systems

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Royal Academy of Engineering, Engineering Leaders Scholar (RAEng ELS) and Mechanical Engineering student at the University of Sheffield with a strong academic background and a passion for the mechanics of creating emotional experiences. Experience with hands-on transit placement experience at Transport for London (TfL DLR). Combining practical SolidWorks mechanism design, life-safety compliance, with leadership and teamwork developed through national ambassador roles, SELSA, volunteering, and interdisciplinary design projects.

Education

University of Sheffield – MEng Mechanical Engineering (Expected Graduation: 2028)

- **Academic Standing:** 1st Class Honours achieved across both Year 1 and Year 2.
- **Honors & Scholarships:** Royal Academy of Engineering Leaders Scholar (RAEng ELS), Sheffield Engineering Leadership and Service Award (SELSA).
- **Relevant Coursework:** Mechanics of Solids & Stress Analysis, Kinematics & Dynamics, Machine Elements, Fluid Dynamics, Thermodynamics, Mechatronics & Control, Materials Science & Manufacturing, MATLAB & Computational Engineering.

Birchwood High School – Bishop's Stortford (2017 – 2024)

- **A-Levels:** Maths (A*), Further Maths (A), Physics (A), German (B) (2024) — Intermediate German Proficiency.
- **GCSEs:** (10 Grade 9–8): Including Maths (9), Physics (9), Chemistry (9), Biology (9), German (9), Geography (9), Design & Technology (8) (2022).
- **FSMQ (Additional Maths):** Grade A (2022).

Engineering & Industry Placement Experience

Transport for London (TfL) — Docklands Light Railway (DLR) – Mechanical Engineering Summer Intern (Summer 2026)

- Worked on station lighting requirements across the network and built heating and cooling calculation models for station and depot building renovations.
- Gained valuable practical perspective on preventive maintenance, life-safety standards, and mechanical building services that keep London moving safely.

Themed Entertainment & CAD Engineering Projects (see portfolio)

Smoke & Starlight AGV — Omnidirectional Trackless Dark Ride – SolidWorks 3D & Python (2025 – 2026)

- Designed an omnidirectional trackless ride chassis in SolidWorks featuring independent dual swerve-drive wheel pods for 360° yaw rotation.
- Programmed trajectory kinematics in Python to verify passenger G-force limits (ASTM F24) and validated chassis structural integrity via FEA (Safety Factor > 3.5).

Time-travel Turntable — Subterranean Spatial Illusion Switch Track – SolidWorks & MATLAB (2025 – 2026)

- Engineered a motorized rotary switch track mechanism utilizing a slewing ring bearing and positive shot-pin locking system for seamless track re-alignment.
- Calculated torque and rotational inertia in MATLAB to achieve smooth 180° rotation in under 4.0s with fail-safe mechanical interlocks (FMEA).

The Smugglers Mine Train — Articulated 3-Wheel Coaster Bogie – SolidWorks & FEA (2025 – 2026)

- Designed an articulated 3-wheel coaster bogie cluster (road, side-friction, upstop wheels) with dual-axis spherical knuckles to eliminate rail-binding.
- Calculated beam bending, pin shear, and dynamic wheel load distributions under maximum G-load cases to EN 13814 / ASTM F24 standards.

Personal Engineering Project — Mini Golf Hole Design & Build – Fabrication & Mechanism Design (Summer 2025)

- Independently designed and constructed a functional mini golf hole with hazards and a rotating windmill obstacle.
- Applied principles of mechanical design, material selection, and basic fabrication techniques using hand tools and simple mechanisms to create interactive features.

Leadership & University Projects

University Group Projects — Mechanical Engineering – University of Sheffield (September 2024 – present)

- **Reaching Aid Design:** Collaborated in a team to develop and present a mechanical reaching aid for individuals with additional needs; managed project documentation, meeting minutes, and team scheduling.
- **Water Bottle Rocket & Launcher:** Designed and tested a water-propelled rocket with custom launcher; analysed trajectory data to predict launch performance and managed group communications.
- **Gearbox Design Project:** Sized and modelled a multi-stage reduction gearbox; calculated gear ratios, shaft stresses, and bearing loads to achieve target torque transmission.

SELSA (Sheffield Engineering Leadership and Service Award) – Faculty of Engineering (September 2025 – Present)

- Selected to join SELSA, a competitive leadership programme for high-achieving engineering students at the University of Sheffield.
- Participating in workshops, industry networking, and team-based challenges focused on leadership, innovation, and social impact.
- Leading the Vulcan to the Sky trust service project based around creating a stand for the Get Up To Speed with STEM fair in March 2026.
- Hackcessible Co-curricular multidisciplinary project focused on increasing accessibility in para ice hockey.

Beaver Leader — 36th Sheffield Scout Group – Sheffield, UK (November 2025 – Present)

- Supporting weekly activities for 6–8-year-olds, encouraging confidence, teamwork, and curiosity.
- Helping to plan and deliver engaging, safe sessions alongside the leadership team.

National Young Ambassador — England Golf & Golf Foundation – United Kingdom (2022 – 2024)

- Worked with a small group of young people from across the country to represent the voice of junior golf in England.
- Represented junior golf nationally at PGA events and outreach events; volunteered with 'Golf Girls Rocks' initiative at a local club.
- Managed social media content and wrote articles for national publications; attended workshops on improving junior golf.
- Hero's Handshake Award — peer-nominated national recognition for leadership and youth mentoring.

Cantate Youth Choir – Bishop's Stortford (2014 – 2024)

- Committed ensemble member for 10 years, performing at concerts and national competitions.

Technical & Transferable Skills

- **Engineering Tools:** SolidWorks 3D (Assemblies, Mates, FEA), Autodesk Fusion 360, AutoCAD (basic), MATLAB, Python (basic), Arduino IDE.
- **Mechanical Design:** Dynamic Bogie Kinematics, Mechanism Articulation, First-Principles Stress Sizing, GD&T (BS 8888 / ASME Y14.5), FMEA / Safety Margins, ASTM F24 & EN 13814 Awareness.
- **Practical Skills:** Hands-on prototyping, 3D printing (FDM/SLA), mechanical assembly, design iteration, manual lathe, drill press, basic fabrication.
- **Soft Skills & Leadership:** Teamwork, leadership, communication, organisation, technical drawing, report writing, public speaking, social media management.
- **Languages:** German (Intermediate proficiency – A-Level qualification).